

Job Market Analysis and Recommendation System

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Course: Data Science and Gen AI

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1. Introduction

The job market is constantly evolving due to technological advancements, economic changes, and shifting business needs. This project aims to analyze job market trends using real-world job posting data and build a recommendation system for job seekers.

The primary objectives include identifying high-demand roles, analyzing salary patterns, understanding geographic variations, and predicting future trends in the job market.

2. Dataset Description

The dataset used in this project consists of job postings collected from Upwork. It includes the following key attributes:

- Job Title
- Published Date
- Hourly Rate (Low & High)
- Budget (for fixed-price jobs)
- Country
- Job Type (Hourly or Fixed)

To standardize salary analysis, a new column called "**salary**" was created by combining hourly rates and fixed budgets.

3. Tools and Technologies Used

- Python
 - Pandas
 - Matplotlib
 - Seaborn
 - Scikit-learn
 - Jupyter Notebook
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4. Data Preprocessing

The dataset was cleaned and prepared using the following steps:

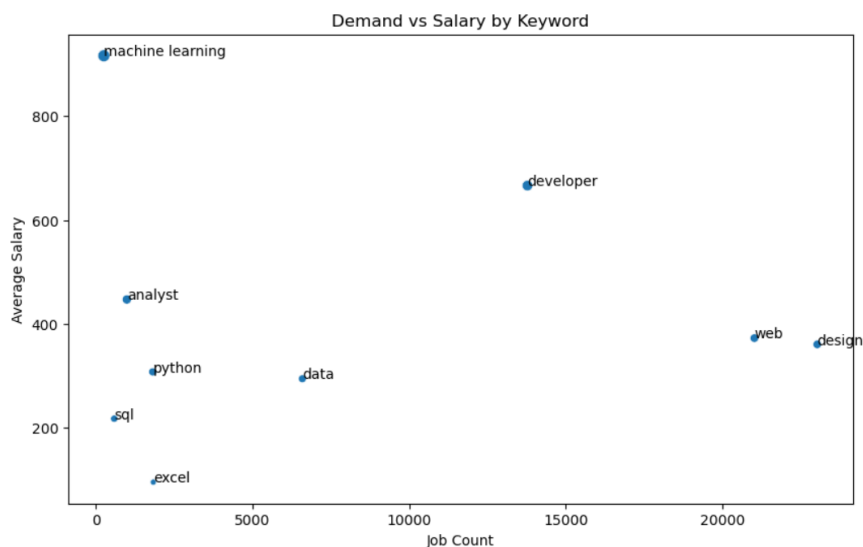
- Converted published dates into datetime format
- Created a unified salary column
- Handled missing values
- Extracted keywords from job titles

5. Task 1: Correlation between Job Titles and Salaries

Job titles were analyzed by extracting key terms such as Python, SQL, Data, and Developer. These keywords were then compared against average salary values.

The analysis revealed that technical roles such as Python and Machine Learning tend to offer higher salaries, while general roles like basic data entry or Excel-based jobs offer comparatively lower compensation.

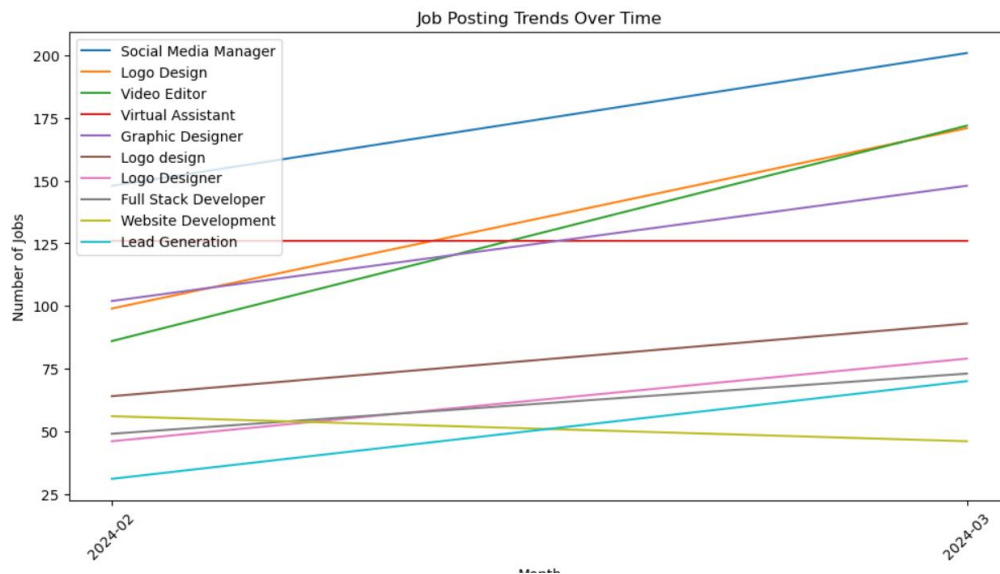
Additionally, job frequency was analyzed alongside salary, highlighting that some roles are high-paying but less frequent, while others are high in demand but offer moderate pay.



6. Task 2: Emerging Job Categories

Job postings were grouped by month to identify trends over time. Frequently occurring roles with consistent presence across months were identified as emerging job categories.

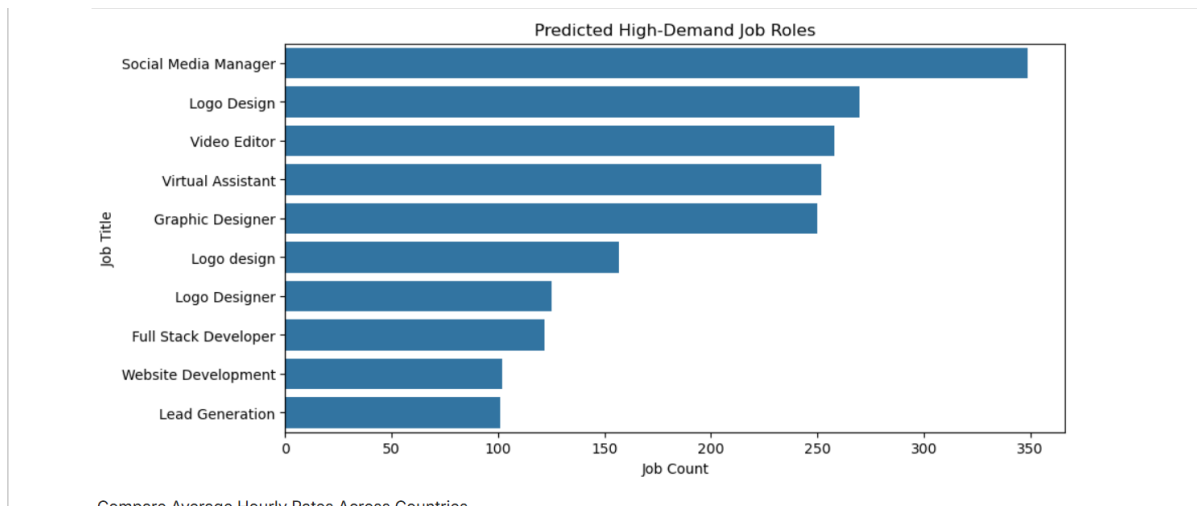
Due to the limited time span of the dataset, the analysis reflects short-term trends. However, it provides useful insights into roles that are currently gaining traction in the job market.



7. Task 3: Predicting High-Demand Job Roles

High-demand roles were predicted by combining job posting frequency and average monthly trends. Roles with consistently high counts and stable presence were identified as likely to remain in demand.

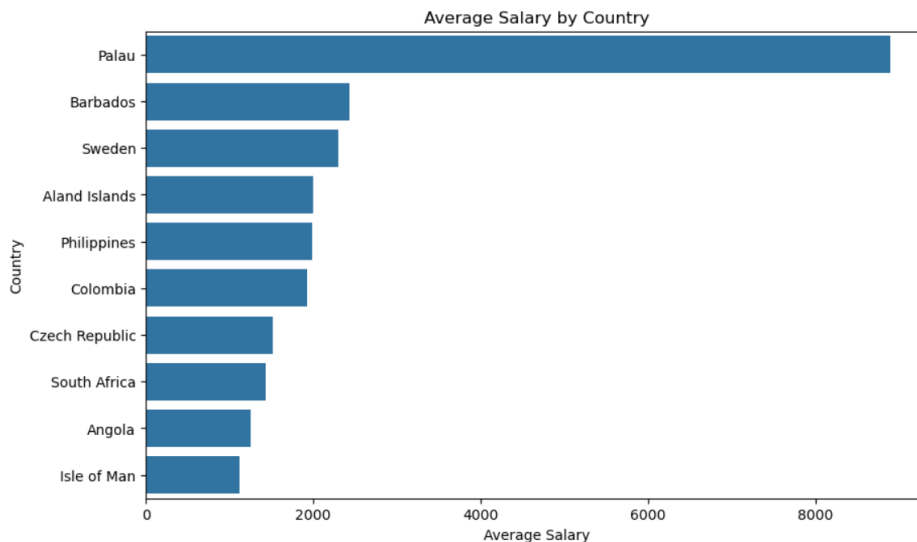
A heuristic approach was used instead of complex machine learning models, ensuring simplicity and interpretability.



8. Task 4: Country-wise Salary Comparison

The average salary across different countries was analyzed to understand geographic variations.

The results showed noticeable differences in compensation levels, which can be attributed to variations in demand, skill availability, and economic conditions. This analysis helps job seekers identify regions with higher earning potential.



9. Task 5: Job Recommendation System

A content-based recommendation system was developed using job titles. TF-IDF vectorization and cosine similarity were used to measure similarity between job titles.

Given an input job title, the system recommends similar roles based on textual similarity. This provides personalized recommendations even in the absence of detailed skill or description data.

10. Task 6: Job Market Trends Over Time

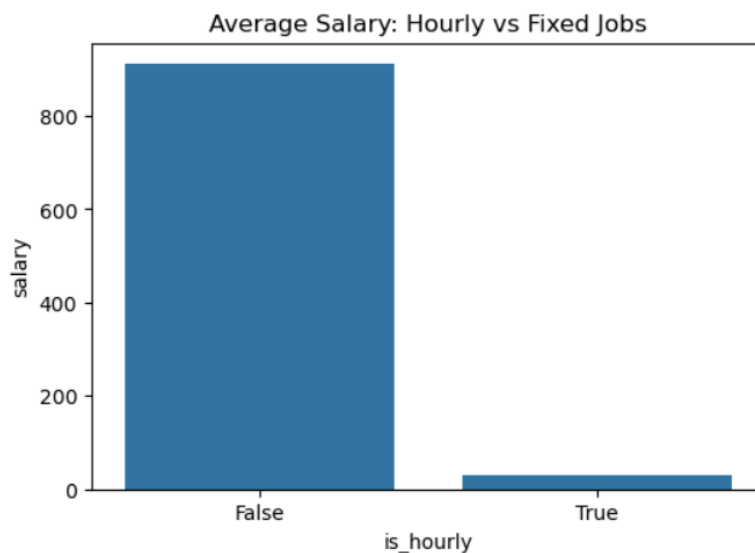
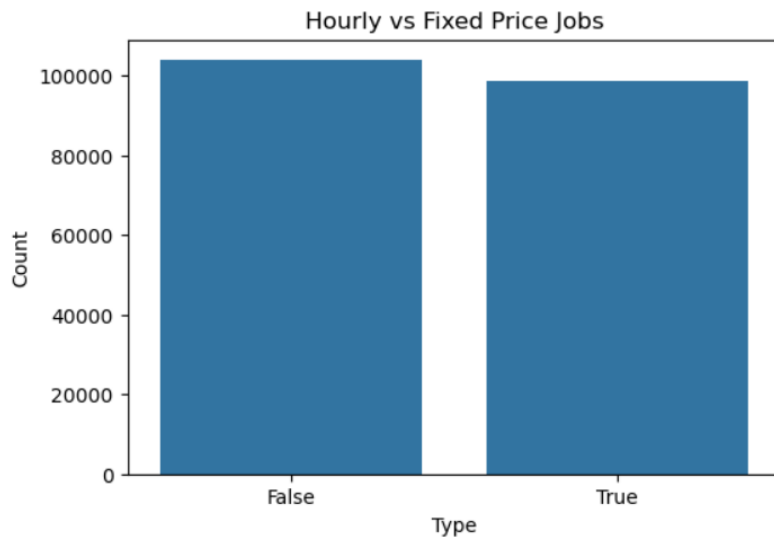
Monthly job posting trends were analyzed to track changes in the job market. Additionally, month-to-month changes were calculated to identify increases or declines in hiring activity.

This helps in understanding short-term fluctuations in job demand.

11. Task 7: Remote Work Trends

Since the dataset primarily consists of remote freelance jobs, the analysis focused on job types (hourly vs fixed-price).

The results show a balance between flexible hourly work and project-based fixed pricing. Additionally, salary comparisons between these job types provide insights into earning potential across different work structures.



12. Task 8: Future Job Market Trends

Future trends were estimated using recent job posting patterns. A simple projection approach was applied using average changes and moving averages to estimate future job demand.

Roles currently experiencing high demand are expected to remain relevant in the near future. These predictions represent short-term forecasts due to limited historical data.

13. Dashboard and Visualization

A visual dashboard was created using Matplotlib and Seaborn to represent trends, salary comparisons, and job demand patterns. These visualizations help in better understanding and interpreting the data.

A visual dashboard was created using Python visualization libraries such as Matplotlib and Seaborn. The dashboard includes charts representing salary trends, job demand, and country-wise comparisons, enabling effective monitoring of job market dynamics.

14. Docker and Deployment

Docker can be used to containerize the application, ensuring consistent deployment across different environments. A Dockerfile and Docker Compose setup can be implemented for scalable and efficient deployment.

Docker can be used to containerize the application for scalable and consistent deployment. A Dockerfile and Docker Compose setup can be implemented to manage dependencies and ensure smooth execution across different environments.

15. Conclusion

This project provides a comprehensive analysis of job market trends using real-world data. Key insights include identification of high-demand roles, salary variations, and emerging job categories.

The recommendation system enhances user experience by suggesting relevant job roles, while trend analysis and forecasting provide valuable insights into the evolving job market.

Overall, this project demonstrates how data analysis and simple machine learning techniques can be used to extract meaningful insights and support decision-making.

Note: The analysis is based on a limited dataset and reflects short-term trends.